NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503
(An Autonomous Institution, Affiliated to Anna University – Chennai)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING &
DEPARTMENT OF INFORMATION TECHNOLOGY

23CS55C / 23IT52C ARTIFICIAL INTELLIGENCE
CO1 Question Bank - 2026-2027 ODD Semester

<table>
<tr><th>Q.no</th><th>Question(Marks)</th><th>CO-CDL</th></tr>
<tr><td>1.</td><td>Specify the PEAS description for the following. (10M)
a) ATM Machine b) Email Spam Filter</td><td>CO1-CDL1</td></tr>
<tr><td>2.</td><td>Schematically discriminate the functioning of the following. (10M)
(a) Goal based Agent b) Utility-based Agent.</td><td>CO1-CDL1</td></tr>
<tr><td>3.</td><td>Consider the following board state: (16M)
<pre>
 X | | O

   | X |

 O | |
</pre>
1.Define the problem formulation for this game.
2.List all legal operators.
3.Generate all possible successor states for Player X.
4.Expand the game tree for the next 3 stages by alternating between Player X and Player O.
5.Mark all terminal states (Win/Loss/Draw).</td><td>CO1-CDL2</td></tr>
<tr><td>4.</td><td>A. ‘Curiosity’ is a Mars rover developed by the United Nations to explore the planet. Identify the type of this intelligent agent and environment. Sketch and explain the working of the agent in its environment with pseudocode. (10)
B. The 8-puzzle problem has an initial state (1,3,4,2,6,7,5,8,blank). Formulate the AI problem and determine whether the goal can be reached from this state. Justify your answer.(6)</td><td>CO1-CDL1</td></tr>
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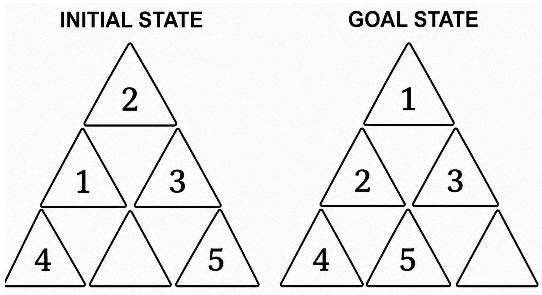
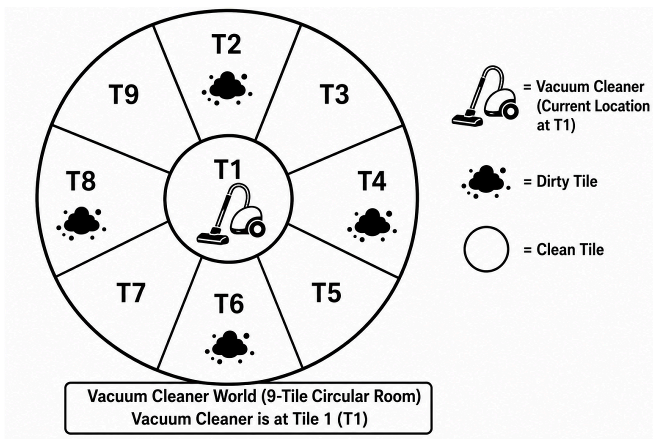
<tr><td>5.</td><td>

An AI agent is assigned with the task of moving the triangles from initial state to goal state. The agent can swap the blank triangle LEFT, RIGHT and DIAGONALLY. Formulate the AI problem, frame the state space up to two levels and solve it using the BFS strategy. (16)</td><td>CO1-CDL2</td></tr>
<tr><td>6.</td><td>

A Vacuum Agent is initially at T1. The agent has to visit all 9 rooms and verify the cleanliness of each room. The rationality of the agent is measured by saving energy in actions taken. Formulate the problem and apply appropriate search strategy to make the agent clean all rooms in an efficient manner. Also return the sequence of actions taken.(16)</td><td>CO1-CDL2</td></tr>
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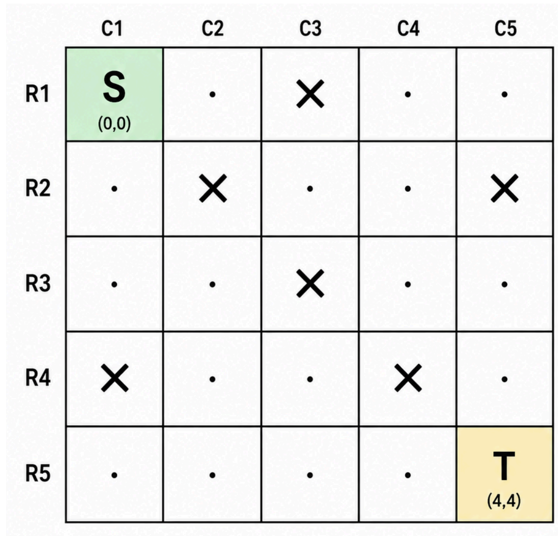

In the Maze problem, ‘S’ represents the initial state and ‘T’ represents the Treasure in the maze. The agent has to find the shortest path to sneak the treasure by moving Left/Right/Down/Diagonally. Explain the Search strategy that can be applied to the agent to reach its goal. Also derive the sequence of actions taken.(16)
</td>
<td>CO1-CDL2</td>
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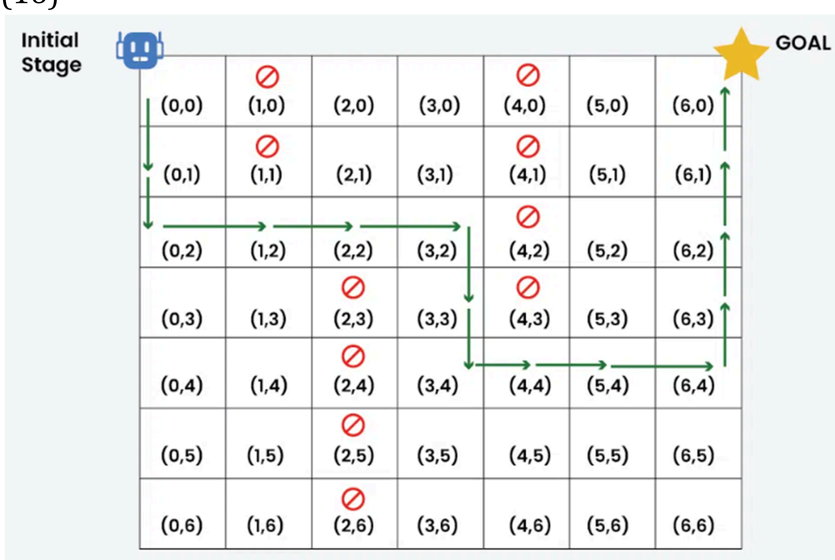
<td>8.</td>
<td>
Given Maze as Grid with Initial Stage (0, 0) & Goal Stage (6, 0). Find the optimal path using a level-ordered search strategy to reach the Goal by exploring child cells with L/R/U/D operations from any grid cell at location (i, j) . Develop a Pseudo code based algorithm for the solution.(16)


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<td>CO1-CDL2</td>
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<td>9.</td>
<td>a) Imagine a bidirectional search scheme where both forward search from the start and backward search from the goal use Breadth-First Search. If the goal is at depth d and the branching</td>
<td>CO1-CDL1</td>
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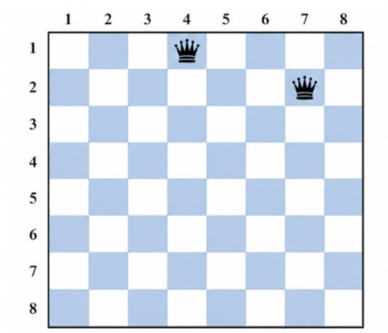
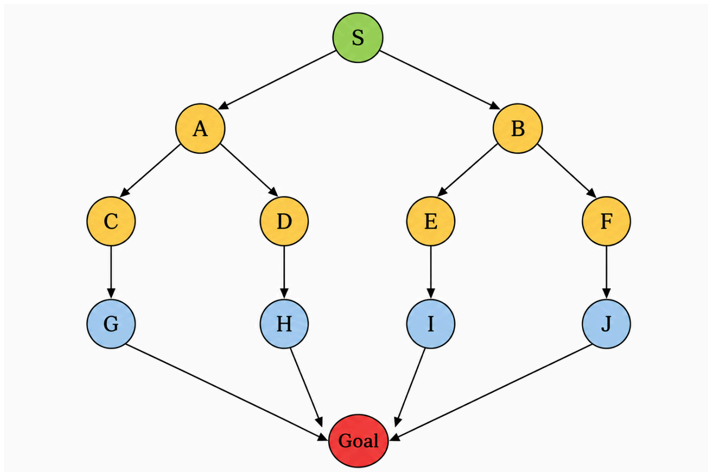
	factor in both directions is b, what is the time complexity of this bidirectional approach? Justify. (4) b) In a graph where edge costs can vary, let g(n) be the cost of the path from the start node to node n. During Uniform Cost Search, when is a node guaranteed to have its absolute shortest path found? Justify. (4)	
10.	An AI agent is assigned the task of solving the 8-Queens Problem. The agent places one queen in each row of an 8 × 8 chessboard such that no two queens attack each other (i.e., no two queens share the same row, column, or diagonal). The current state of the chessboard, generated after placing the first two queens, is shown in the figure below. The remaining empty cells are available for placing the other queens.  Using the given board, formulate the AI problem by specifying the initial state, state space, goal state and path cost. Apply an appropriate search strategy to obtain a valid sequence of queen placements that reaches the goal state. Draw the final board configuration and justify why it is a valid solution. (16M)	CO1-CDL2
11.	Consider the Missionaries and Cannibals problem. Three missionaries and three cannibals are on the left bank of a river with one boat that can carry a maximum of two people. At no point should missionaries be outnumbered by cannibals on either bank. (a) Formulate the problem by defining the initial state, goal state, operators, and constraints. (6M) (b) Draw the state-space tree up to three levels. (4M) (c) Apply Depth-First Search (DFS) to determine a valid sequence of moves to reach the goal state. (6M)	CO1-CDL2
12.	An autonomous warehouse robot must move from the Start (S) location to the Goal (G) while avoiding obstacles in a 6 × 6 grid. The robot can move Up, Down, Left, and Right. (a) Formulate this as an AI search problem by specifying the initial state, state space, actions, goal test, and path cost. (6M)	CO1-CDL2

	(b) Construct the search tree up to two levels. (4M) (c) Apply Uniform Cost Search (UCS) to determine the optimal path and list the sequence of actions. (6M)	
13.	Consider the following graph where S is the start node and move to Goal state.  (a) Apply Depth-First Search (DFS) and show the order of node expansion. (6M) (b) Apply Breadth-First Search (BFS) and determine the shortest path from S to Goal. (6M) (c) Compare BFS and DFS in terms of completeness, optimality, time complexity, and space complexity. (4M)	CO1-CDL2
14.	Consider the Water Jug Problem with a 4-litre jug and a 3-litre jug. Initially, both jugs are empty. The objective is to obtain exactly 2 litres in the 4-litre jug. (a) Formulate the AI problem by specifying the initial state, goal state, operators, and state space. (6M) (b) Draw the state-space tree up to three levels. (4M) (c) Apply Breadth-First Search (BFS) to obtain the solution path.(6M)	CO1-CDL2
15.	Consider the Farmer, Fox, Goose, and Grain river crossing problem. A farmer must transport a fox, a goose, and a bag of grain across a river using a boat that can carry only the farmer and one additional item. The fox cannot be left alone with the goose, and the goose cannot be left alone with the grain. (a) Formulate the problem by specifying the initial state, goal state, operators, and constraints. (6M) (b) Draw the state-space tree up to three levels. (4M) (c) Apply Breadth-First Search (BFS) to derive a valid sequence of crossings. (6M)	CO1-CDL2